Software Engineering & Project

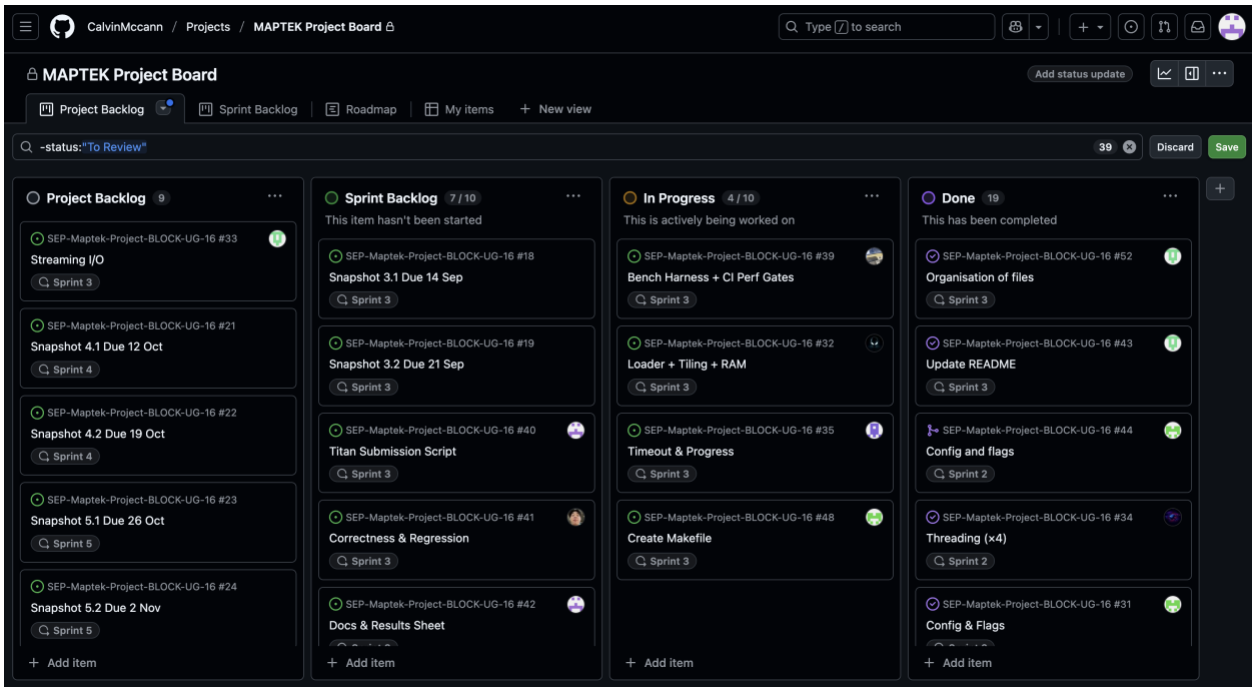
Snapshot week 7 - BLOCK UG-16

Maptek, Block Model Compression Algorithm

Name	Student ID
Adam Dao	a1797499
Calvin McCann	a1860428
Hamidur Rahim Mahir	a1914792
Izak Stentiford	a1822100
Mohammed Ali Rezaei	a1714717
Pezhman Beheshtian	a1661736
Tyler McKinley	a1716769
Yuzhe Ma	a1918429

Product Backlog and Task Board

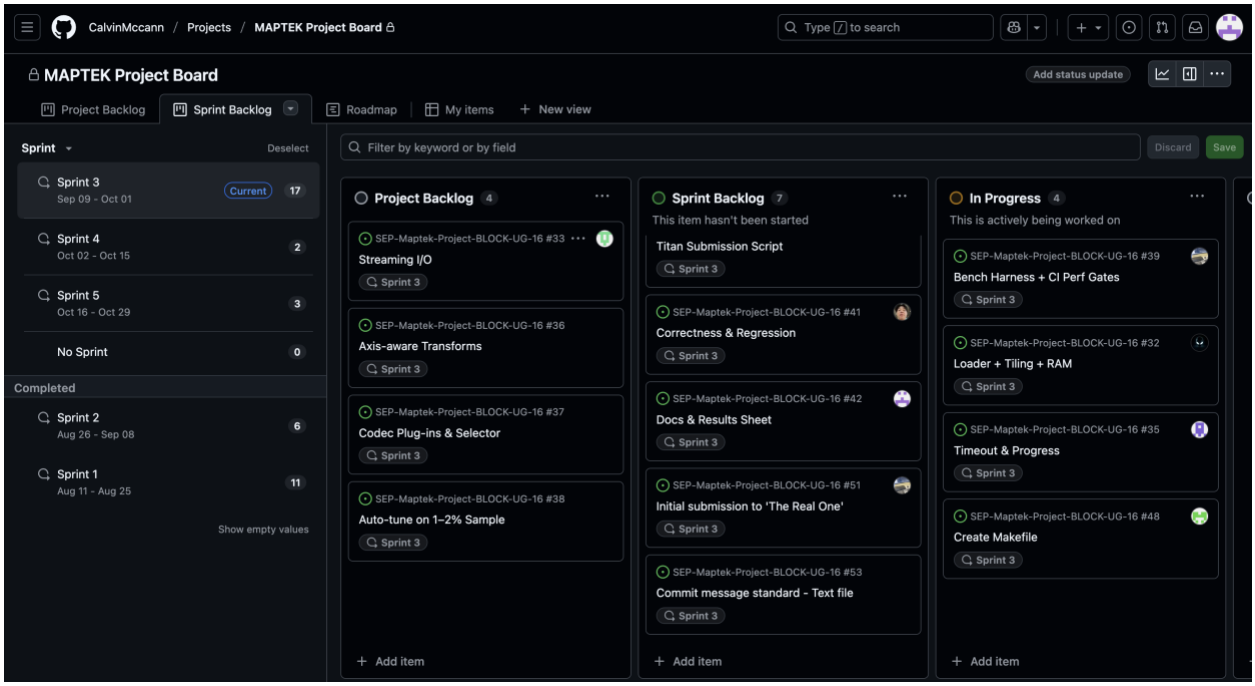
Product Backlog	
Task	**Description**
Axis-aware Transforms	delta/RLE per axis Palette/dictionary per tile
Snapshot 3.1	Create snapshot 3.1
Snapshot 3.2	Create snapshot 3.2
Snapshot 4.1	Create snapshot 4.2
Snapshot 4.2	Create snapshot 4.2
Snapshot 5.1	Create snapshot 5.1
Snapshot 5.2	Create snapshot 5.2
Correctness and Regression	Do: run current code on intro and fast competitions; confirm outputs and acceptance still pass. Done when: both older comps pass their acceptance checks.
Docs & Results Sheet	Shared results sheet (commit, params, ratio, MB/s, RAM, runtime)
Final Presentation	Initial planning for final presentation



Sprint Backlog and User Stories

Sprint Backlog	
Task	**Description**
Docs & results sheet	Shared results sheet (commit, params, ratio, MB/s, RAM, runtime)
Correctness & regression	Do: run current code on Intro and Fast competitions; confirm outputs and acceptance still pass. Done when: both older comps pass their acceptance checks.
Sprint retrospective 2	Create sprint retrospective 2
Streaming I/O	prototype buffered read, mmap, and prefetch thread; time each; keep the fastest path under 8GB.
Snapshot 2.2	Create snapshot 2.2
Create makefile	Simplify the compilation process
Timeout & progress	global wall-clock timer; at 80m either exit cleanly or switch to “fast mode”; write summary stats before exit.

Sprint Backlog	
Task	**Description**
Organisation of files	Separate files by challenge. e.g. The RealOne.cpp, TheFastOne.cpp, etc. cleaner structure.



User Stories (Sprint 3)

For a user of the block model compression algorithm, the algorithm needs to be optimised for speed, rigorously tested, and supported by more organised workflows so that larger datasets can be managed efficiently, and results trusted. In this sprint, the team will focus on improving performance by refining existing working code and reducing inefficiencies, while also implementing more structured testing to validate correctness and reliability across a wider range of block model scenarios.

To ensure these improvements are sustainable, workflows and project structure have been made cleaner and more consistent, making collaboration smoother and reducing the likelihood of errors. Functional algorithms from previous sprints will be revisited and polished to improve clarity, maintainability, and performance. As the Product Owner (TA), I have reinforced the importance of balancing optimisation with stability and worked with the team to set realistic goals that keep progress aligned with project deadlines and quality standards.

Definition of Done

Optimisation of the working compression algorithm such that it performs well on 'The Real One' compression scenario and works faster and more optimally on previous challenges.

Summary of Changes

This week was defined by organisational changes to the product and sprint backlog. This allowed for a more simplistic and efficient management of tasks and deliverables. One component of this was dividing the task board into specific sprint categories, allowing easier viewing of tasks in previous, current and future sprints.

Having had several meetings throughout the week, the team has allocated tasks and roles for sprint 3 that will progress the project to achieve targets by the end of sprint 3. The team has agreed that more effective testing should be a priority as well as optimisation of the working compression algorithm.